

E-Commerce Data Analysis using SQL

Project Overview

This project demonstrates an end-to-end SQL Data Analysis workflow using an E-Commerce dataset. The objective is to transform raw imported data into meaningful business insights through data quality assessment, validation, exploratory analysis, KPI reporting, and advanced SQL techniques.

The project follows a real-world analytical workflow similar to that used by Data Analysts in organizations. Rather than simply writing SQL queries, the project emphasizes data quality, business reporting, reusable database objects, and analytical thinking.

Business Objective

The primary objective of this project is to analyze customer purchasing behavior, product performance, sales trends, and business performance using SQL.

The analysis aims to answer questions such as:

- Which products generate the highest revenue?
 - Which customers contribute the most revenue?
 - Which categories perform the best?
 - Which states and cities generate the highest sales?
 - What is the overall revenue after discounts?
 - Which payment methods are most popular?
 - What are the monthly and yearly sales trends?
 - What KPIs should management monitor regularly?
-

Dataset Description

The project uses three relational tables imported from an Excel workbook.

Customers Table

Contains customer information including:

- Customer ID
- Customer Name
- Gender
- City

- State
 - Signup Date
 - Membership
 - Age
-

Products Table

Contains product information including:

- Product ID
 - Product Name
 - Category
 - Brand
 - Price
 - Stock Quantity
 - Supplier
 - Rating
-

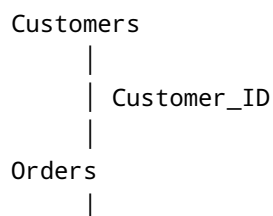
Orders Table

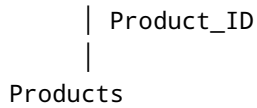
Contains order transaction information including:

- Order ID
 - Customer ID
 - Product ID
 - Order Date
 - Quantity
 - Discount
 - Payment Method
 - Order Status
-

Database Schema

The project consists of three primary tables.





```
graph TD; Product_ID --- Products;
```

Relationships:

- One Customer → Many Orders
- One Product → Many Orders

The Orders table acts as the transaction table connecting Customers and Products.

Project Workflow

The SQL project is divided into multiple phases.

Phase 1 — Database Setup & Initial Data Inspection

Tasks performed:

- Created database
 - Imported Excel sheets
 - Verified imported tables
 - Inspected table structure
 - Counted rows and columns
 - Verified data types
-

Phase 2 — Data Quality Assessment

Performed comprehensive quality checks including:

- NULL values
 - Blank values
 - Duplicate rows
 - Duplicate IDs
 - Invalid values
 - Inconsistent text values
 - Referential integrity
 - Outlier detection
-

Phase 3 — Data Standardization

Standardized the dataset by:

- Removing leading/trailing spaces
 - Converting blank strings into NULL
 - Standardizing text values
 - Standardizing category names
 - Standardizing membership values
 - Standardizing payment methods
 - Creating backup tables before modifications
-

Phase 4 — Data Validation

Validated data after standardization by checking:

- Missing values
 - Duplicate IDs
 - Invalid age
 - Invalid price
 - Invalid quantity
 - Invalid rating
 - Invalid discount
 - Invalid foreign keys
 - Future dates
 - Accepted categorical values
-

Phase 5 — Exploratory Data Analysis (EDA)

Explored the dataset using SQL by analyzing:

- Customer distribution
 - Membership distribution
 - Product distribution
 - Category distribution
 - Brand distribution
 - Supplier distribution
 - Payment methods
 - Order status
 - Monthly orders
 - Yearly orders
 - Basic descriptive statistics
-

Phase 6 — Business Analysis

Business-focused SQL analysis including:

- Total Revenue
 - Net Revenue
 - Discount Analysis
 - Average Order Value
 - Revenue by Category
 - Revenue by Brand
 - Revenue by Customer
 - Revenue by Membership
 - Revenue by State
 - Revenue by City
 - Product Performance
 - Monthly Revenue
 - Yearly Revenue
-

Sales Summary View

A reusable reporting view named **sales_summary** was created to avoid repeatedly joining multiple tables.

The view combines:

- Customer information
- Product information
- Order information
- Gross Revenue
- Discount Amount
- Net Revenue

This simplified later KPI reporting and advanced SQL analysis.

Phase 7 — KPI Analysis

Key Performance Indicators calculated:

Sales KPIs

- Gross Revenue
- Discount Amount
- Net Revenue
- Average Order Value
- Total Orders

- Total Quantity Sold

Customer KPIs

- Total Customers
- Repeat Customers
- Revenue per Customer

Product KPIs

- Products Sold
- Average Selling Price
- Average Quantity per Order

Operational KPIs

- Cancellation Rate
- Revenue by Membership
- Revenue by State
- Revenue by Payment Method
- Monthly Revenue
- Yearly Revenue

Phase 8 — Advanced SQL Analysis

Advanced SQL concepts used:

- Common Table Expressions (CTEs)
- Window Functions
- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- LAG()
- Running Totals
- Revenue Contribution
- Customer Lifetime Value (CLV)
- Top-N Analysis
- Category Ranking

Phase 9 — Business Reporting Objects

Created reusable reporting objects.

Views

- Customer Revenue Summary
- Product Performance
- Monthly Sales Summary
- Category Performance
- State Revenue Summary

Stored Procedures

- Top Customers
- Revenue by Year
- Category Performance

These objects can be directly connected to reporting tools like Power BI or Tableau.

SQL Concepts Used

The project demonstrates practical use of:

- SELECT
 - WHERE
 - ORDER BY
 - GROUP BY
 - HAVING
 - DISTINCT
 - Aggregate Functions
 - CASE Expressions
 - JOINS
 - Subqueries
 - Common Table Expressions (CTEs)
 - Window Functions
 - Ranking Functions
 - Date Functions
 - String Functions
 - Mathematical Functions
 - Views
 - Stored Procedures
-

Business Insights

The project enables management to identify:

- High-value customers
 - Best-selling products
 - Highest revenue categories
 - Strongest performing brands
 - High-performing cities and states
 - Customer purchasing patterns
 - Revenue trends over time
 - Discount impact on revenue
 - Preferred payment methods
 - Membership performance
-

Key Learnings

Through this project, the following skills were developed:

- Relational database design
 - Data quality assessment
 - Data validation
 - SQL data cleaning
 - Business reporting
 - KPI development
 - Advanced SQL querying
 - Window functions
 - Performance reporting
 - Analytical problem-solving
-

Future Improvements

Potential enhancements include:

- Creating indexes to optimize query performance.
 - Implementing triggers for automatic data validation.
 - Adding foreign key constraints and primary key constraints after data quality checks.
 - Building interactive dashboards in Power BI and Tableau using the SQL database.
 - Automating recurring reports using scheduled stored procedures.
 - Expanding the project with customer segmentation (RFM analysis), cohort analysis, and sales forecasting.
-

Conclusion

This project demonstrates a complete SQL-based data analysis workflow, beginning with raw imported data and progressing through data quality assessment, standardization, validation, exploratory analysis, business analysis, KPI reporting, advanced SQL techniques, and reusable reporting objects.

It showcases the practical SQL skills expected from a Data Analyst and provides a strong foundation for building interactive dashboards and making data-driven business decisions.